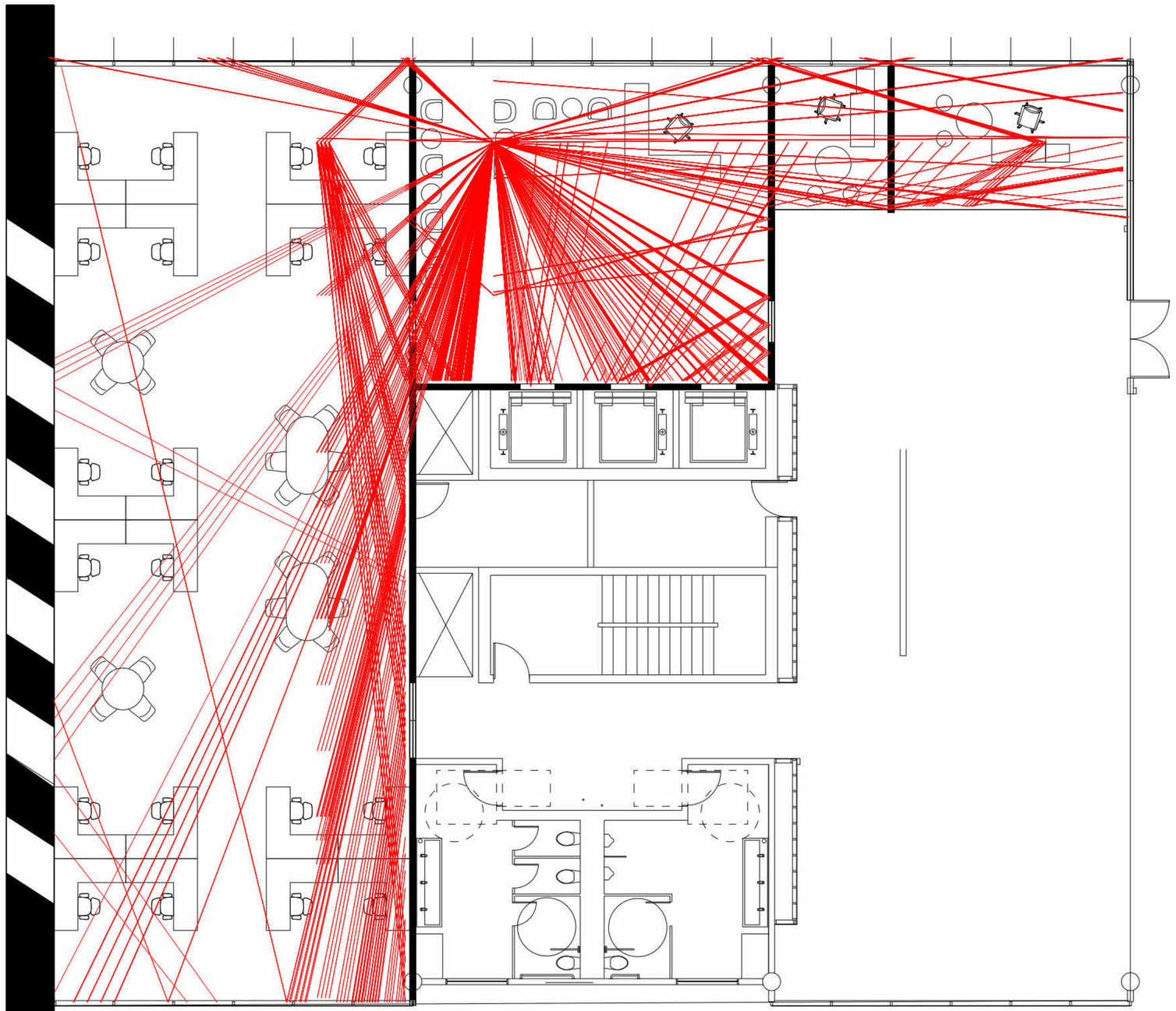


ENLARGED ACOUSTICAL PLAN 1: SOUND ANALYSIS WITH THE SOURCE IN THE RECEPTION AREA.

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No.	Description	Date

CONSTRUCTION DOCUMENT SET

ENLARGED OFFICE PLAN - ACOUSTICAL ANALYSIS

Date	04.08.2024
Drawn By	Author
Checked By	Checker

A502

Scale 1/4" = 1'-0"



ENLARGED ACOUSTICAL PLAN 2: SOUND ANALYSIS WITH THE SOURCE IN THE OPEN OFFICE AREA.

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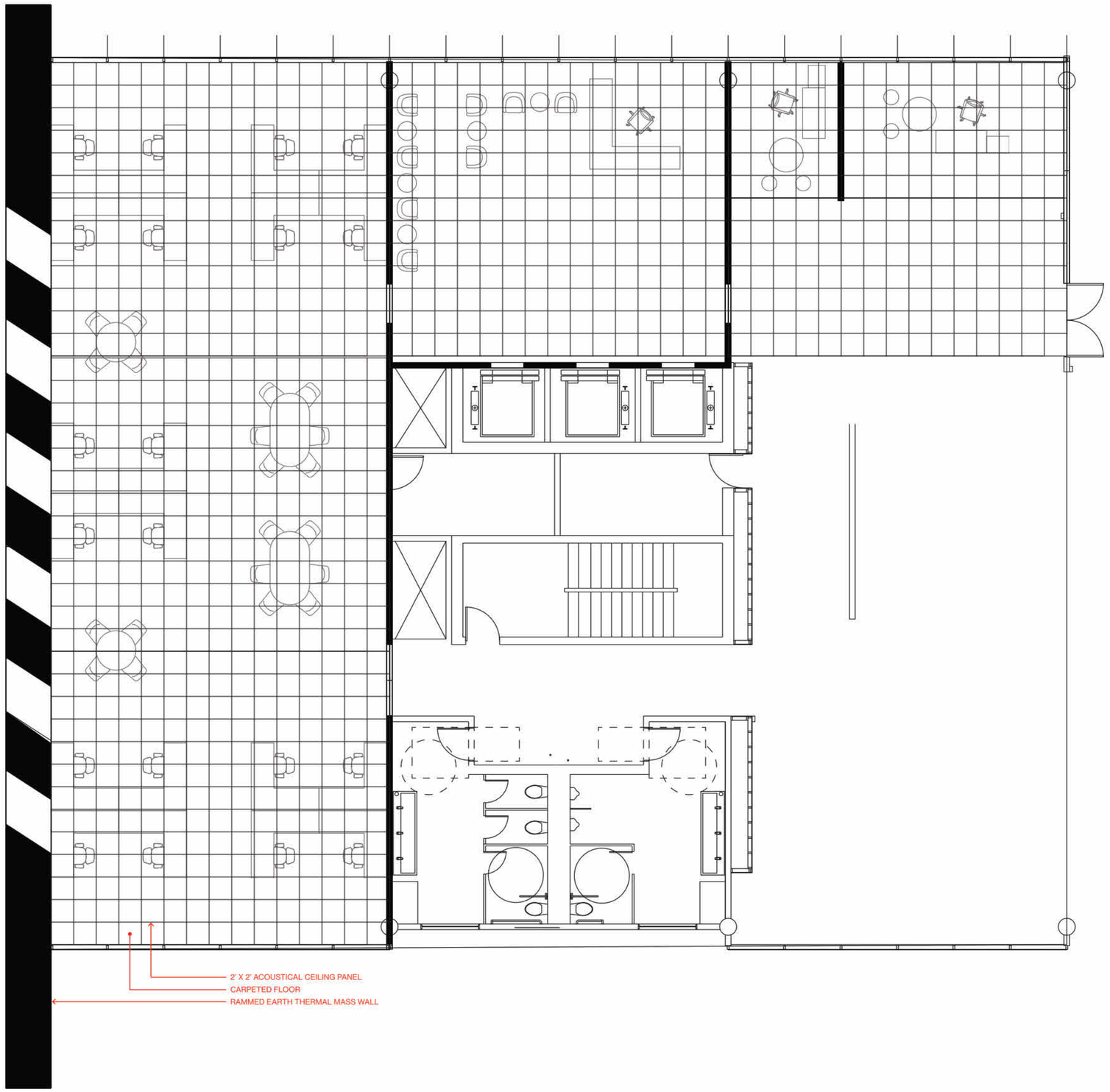
CONSTRUCTION DOCUMENT SET

ENLARGED OFFICE PLAN - ACOUSTICAL ANALYSIS

Date	04.08.2024
Drawn By	Author
Checked By	Checker

A503

Scale 1/4" = 1'-0"



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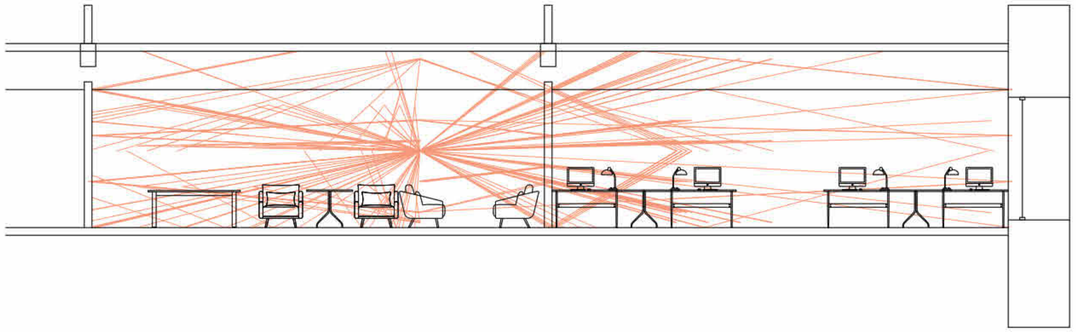
CONSTRUCTION DOCUMENT SET

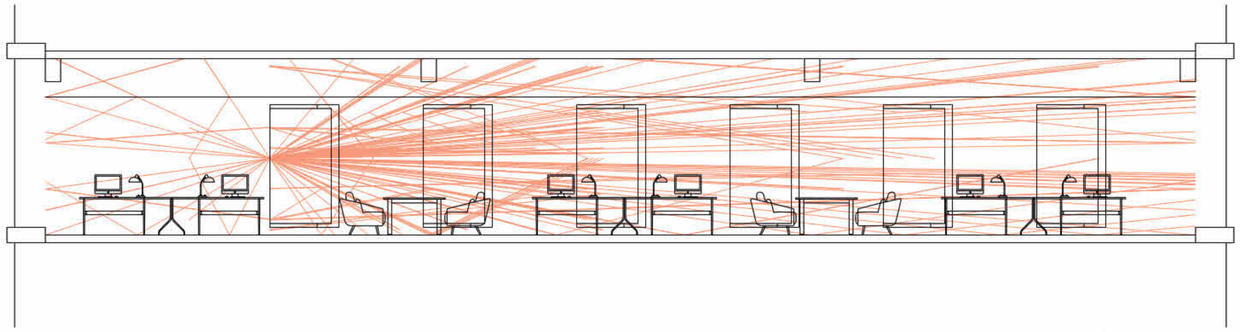
REFLECTED OFFICE CEILING PLAN

Date	04.08.2024
Drawn By	Author
Checked By	Checker

A504

Scale 1/4" = 1'-0"





Sections At 1/4" Scale

ACOUSTICAL CALCULATIONS

Material:
Carpet Floor, Curtain Wall System, Rem Earthed Wall, 6" Concrete Wall, Acoustic Ceiling Panel

Area Calculation
-For Office
Carpet = 2250 sqft
Office Earthern Wall = 75 * 12 = 900 sqft
MEC Room = 25 * 12 = 300 sqft
Bathroom = 19 * 12 = 228 sqft
Eathern Wall = 75 * 12 = 900 sqft
Curtain Wall = 30 * 12 * 2 = 720 sqft
Windows = 4.5 * 8 and 7 = -252 sqft
Doors = 3 * 8 = -24 sqft
Acoustic Ceiling Panel = 2250 sqft

-For Reception-
Carpet = 750 sqft
Walls 18 * 10 and 2 pieces = 360 sqft
MEC Room 30 ft * 12 = 360 sqft
Doors = 3 * 8 * 3 = -72 sqft
Acoustic Ceiling Panel= 750 sqft

Sabin Calculation
-For Office-
Carpet = 2250 * 0.2 = 450
6" Concrete 504 * 0.3 = 151.2
Curtain Wall = 720 * 0.05 = 36
Eathern Wall = 648 * 0.668 = 432.8
Acoustic Ceiling Panel = 2250 * 0.7 = 1575

Total Sabin = 2645.1

-For Reception-
Carpet = 750 * 0.2 = 150
6" Concrete = 648 * 0.3 = 194.4
Curtain Wall = 360 * 0.05 = 18
Acoustic Ceiling Panel = 750 * 0.7 = 525

Total Sabin = 887.4

Volume
V = 27000 for Office and 9000 for Reception

Reverbation Time
T = K * V / S
T = 0.51 for Office and 0.51 for Reception Area (0.5< T < 1.1)

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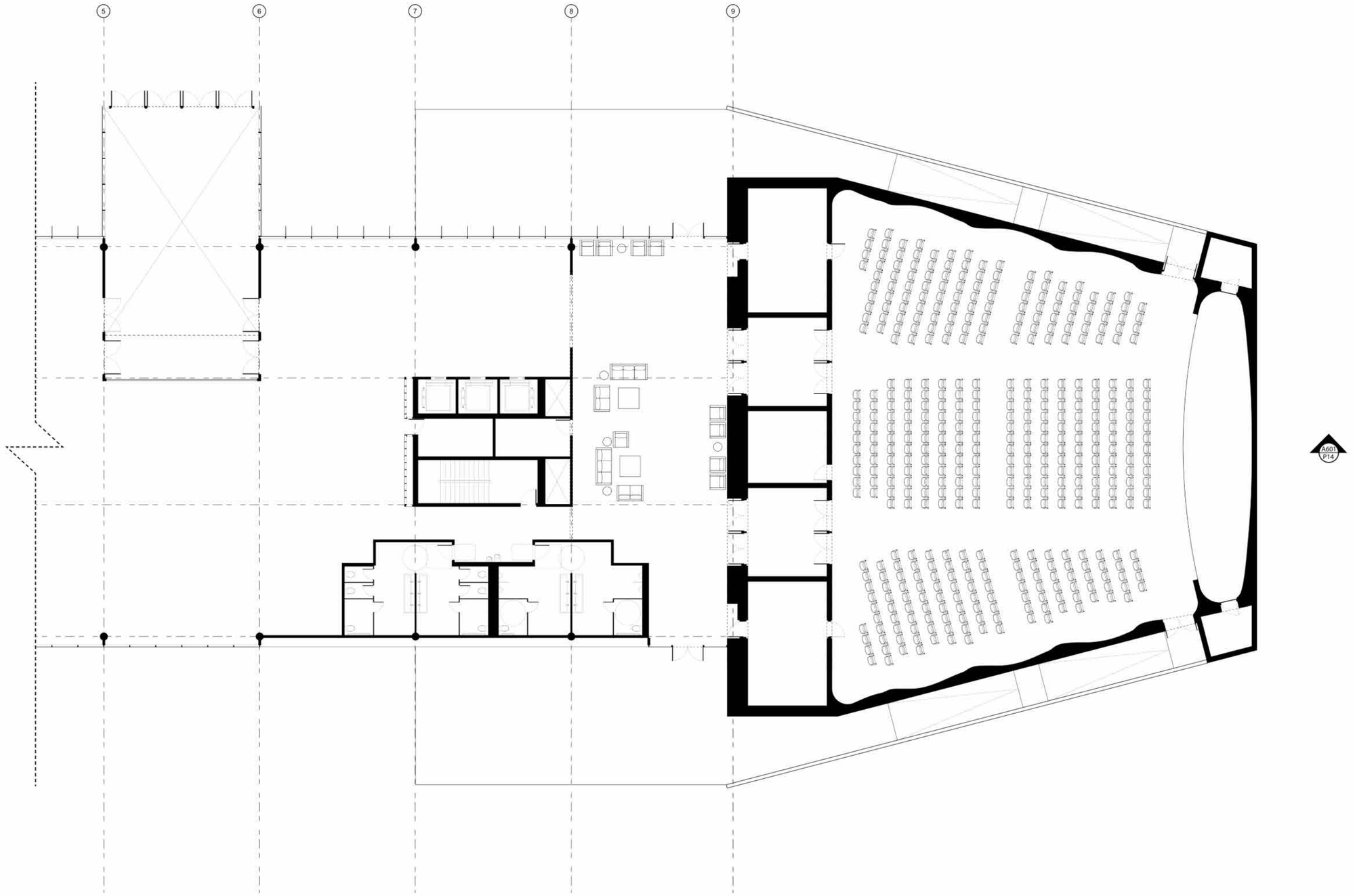
CONSTRUCTION DOCUMENT SET

ENLARGED OFFICE SECTIONS - ACOUSTICAL ANALYSIS

Date	04.08.2024
Drawn By	Author
Checked By	Checker

A505

Scale 1/4" = 1'-0"



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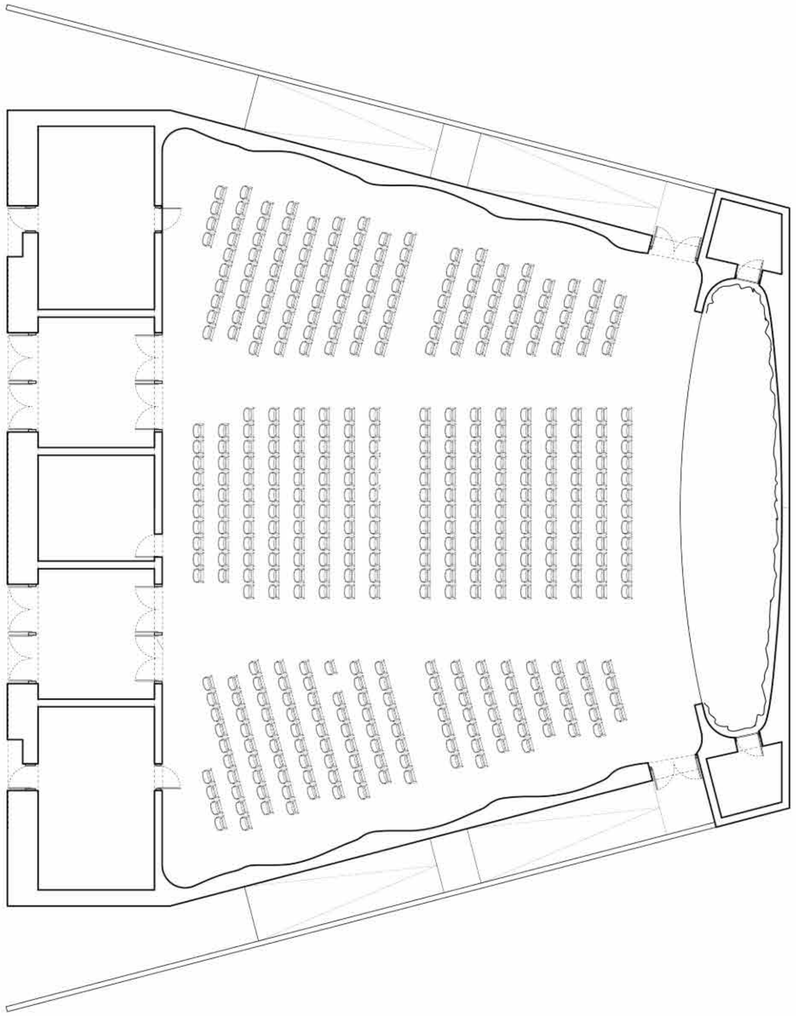
CONSTRUCTION DOCUMENT SET

GROUND FLOOR PLAN - AUDITORIUM

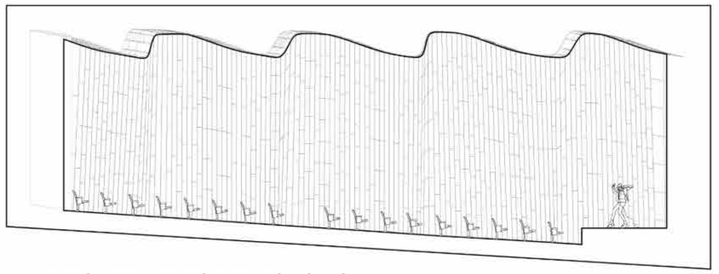
Date	04.08.2024
Drawn By	Author
Checked By	Checker

A600

Scale 1/8" = 1'-0"



ENLARGED AUDITORIUM PLAN



ENLARGED AUDITORIUM SECTION

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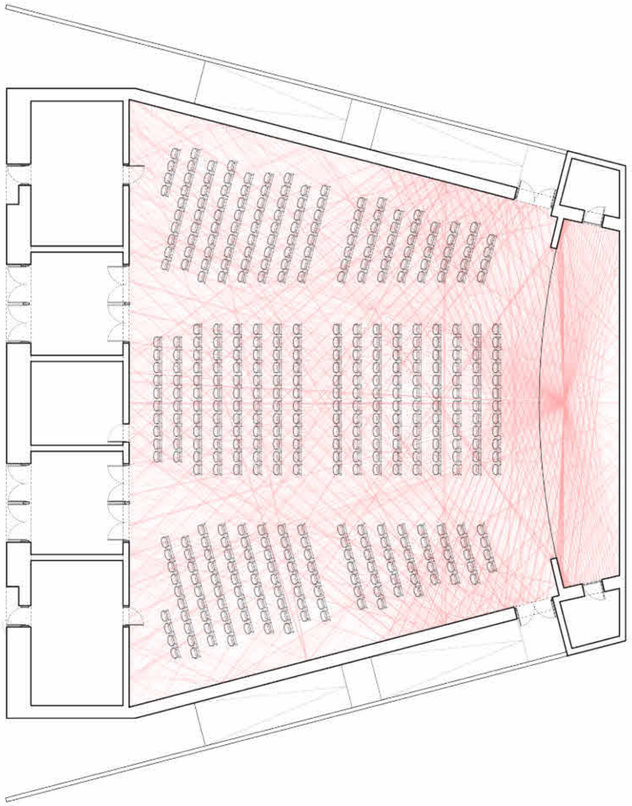
CONSTRUCTION DOCUMENT SET

ENLARGED AUDITORIUM PLAN AND SECTION

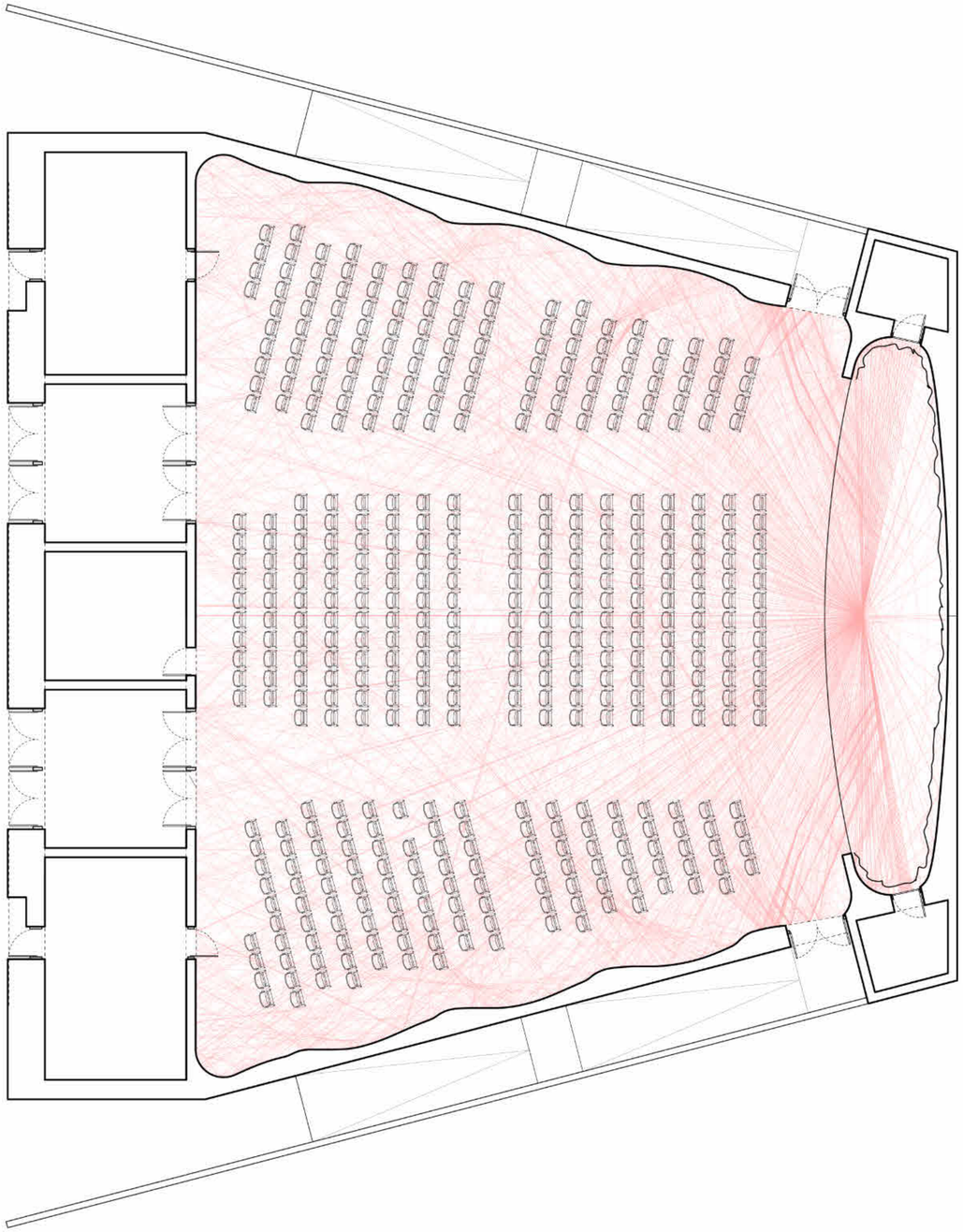
Date	04.08.2024
Drawn By	Author
Checked By	Checker

A601

Scale 1/8" = 1'-0"



BASE AUDITORIUM



AUDITORIUM WITH FINAL ACOUSTIC ALTERATIONS

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CONSTRUCTION DOCUMENT SET

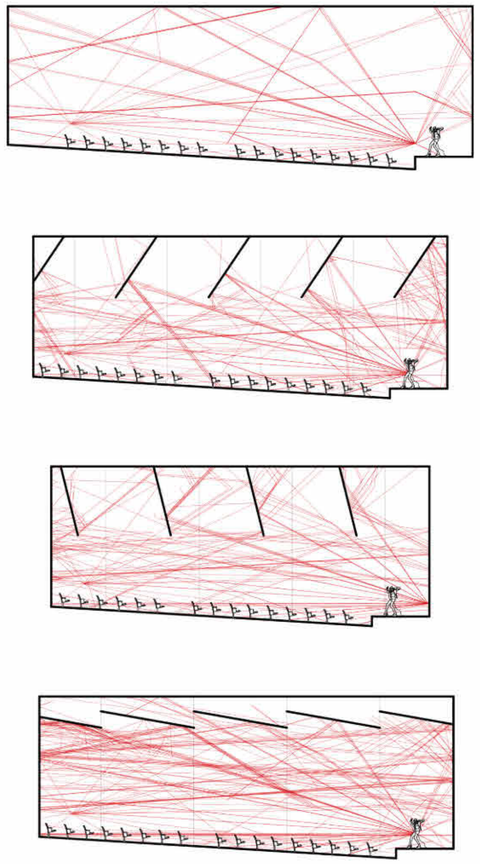
ACOUSTICS ANALYSIS - PLAN

Date	04.08.2024
Drawn By	Author
Checked By	Checker

A602

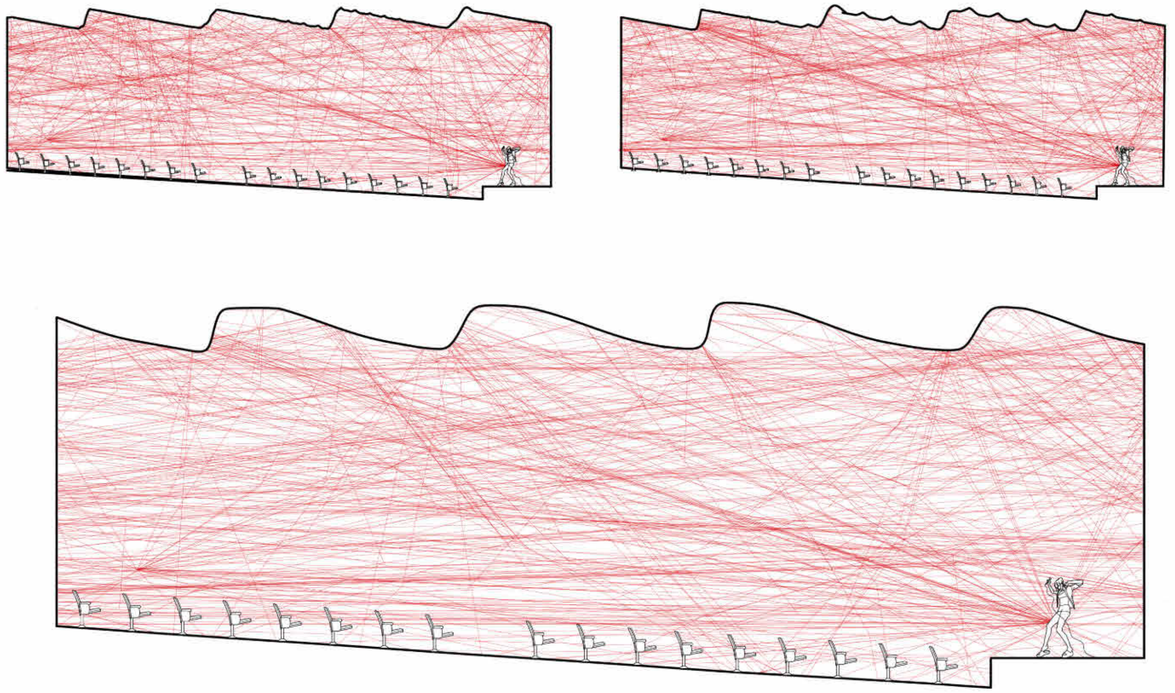
Scale

PROTOTYPE



TESTING DIFFERENT TYPES OF CEILING PANELS TO IMPROVE THE QUALITY OF SOUND IN THE BACK OF THE AUDITORIUM

FORM FINDING



EXPERIMENTING WITH SHAPES OF CURVES TO ENSURE SOUNDWAVES ARE SPREAD MORE EVENLY THROUGHOUT.

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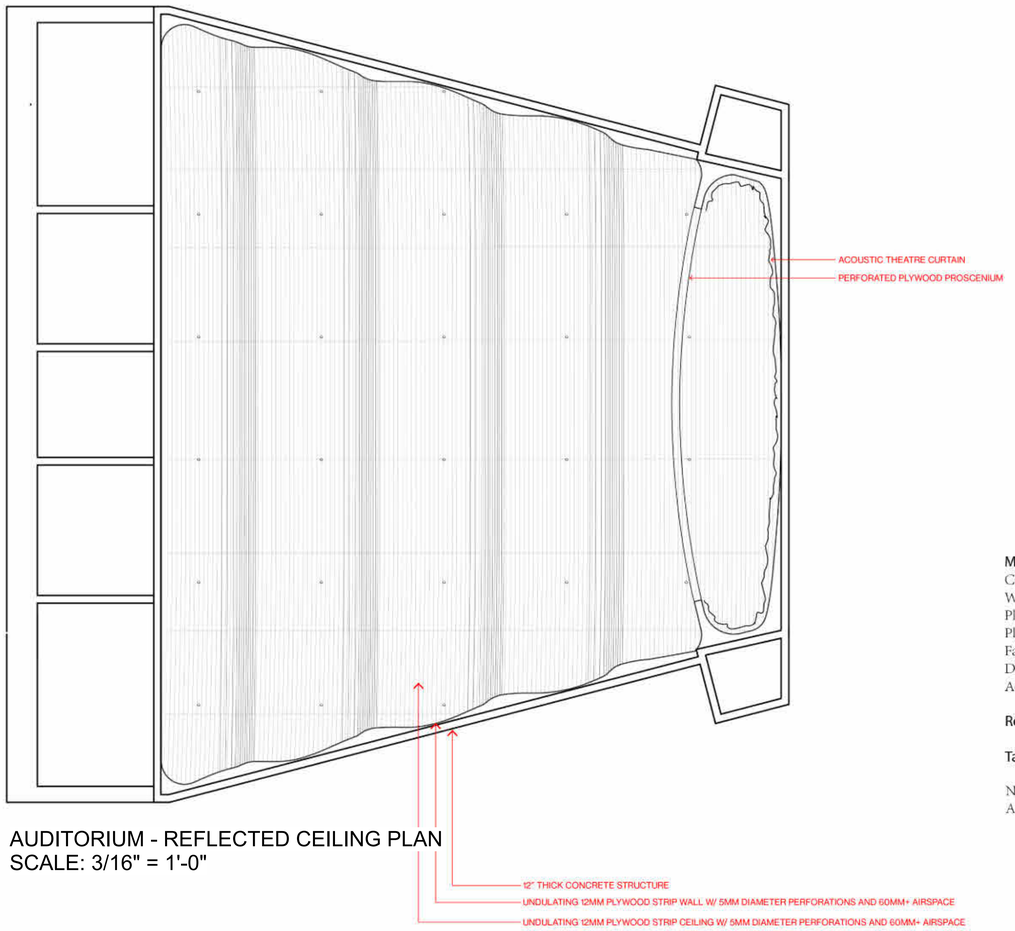
CONSTRUCTION DOCUMENT SET

ACOUSTICS ANALYSIS - SECTION ITERATIONS

Date	04.08.2024
Drawn By	Author
Checked By	Checker

A603

Scale



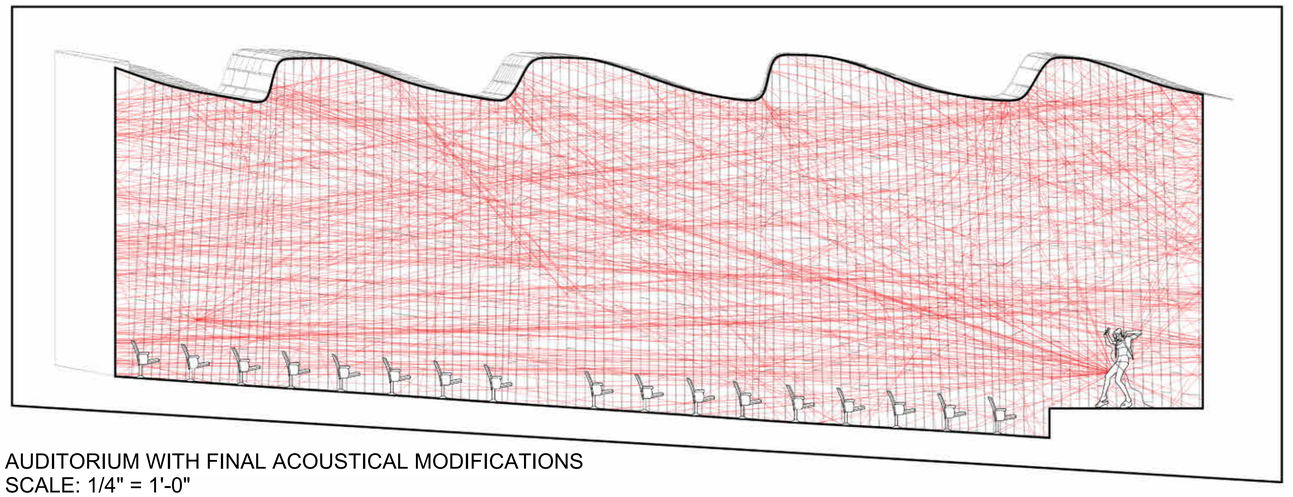
AUDITORIUM - REFLECTED CEILING PLAN
SCALE: 3/16" = 1'-0"

Material	Sq Footage	Absorption Coefficient	Sabins
Carpet - Floor	4660.86	0.15	699.129
Wood platform with large space beneath - Stage	932	0.17	158.44
Plywood 12mm thick perforated 5mm diameter - Ceiling	6265.93	0.3	1879.779
Plywood 12mm thick perforated 5mm diameter - Walls	6489.71	0.3	1903.038
Fabric Upholstered Seating - fully occupied	1714.5	0.96	1645.92
Drapery (18 oz/yd2, 612 g/m2, pleated 50%)	1315.07	0.75	986.3025
Acoustic doors	146.25	0.49	71.6625

Reverb Time =0.05 * (Total Volume / Total Sabins) = **0.05 * (122148.61 / 7344.271) = 0.8315911137**

Target Reverb Range 0.5 - 1.1 seconds

NOTE: ABSORBTION COEFFICIENT FOUND ASSUMING 1,000 HZ
ABSORBTION COEFFICIENT SOURCE: https://www.acoustic supplies.com/absorption-coefficient-chart/
https://www.acoustic.ua/st/web_absorption_data_eng.pdf



AUDITORIUM WITH FINAL ACOUSTICAL MODIFICATIONS
SCALE: 1/4" = 1'-0"

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CONSTRUCTION DOCUMENT SET

ACOUSTICAL ANALYSIS SECTION AND RCP

Date	04.08.2024
Drawn By	Author
Checked By	Checker

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Scale AS NOTED